

What a Zero Readout Certifies

Zero as the failure locus of retained distinction

Yaoharee Lahtee

Open Civil Science Initiative, Pattani, Thailand

ORCID: 0009-0005-3861-0626

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Abstract

Arithmetic supplies the symbol 0. It does not, by itself, say what a measurement returning that symbol has established. This paper answers that question for a setting whose primitive is the retained difference—the assertion that two things are told apart under finite access—and whose readings are finite and rational. In such a setting zero is anomalous: it arrives in the same register as every other value, yet reports that the primitive did not fire. We show that a zero readout is not an absence but a classification, and prove which one. For the retained-information operator on a positively weighted comparison graph, the zero locus coincides, in both directions, with the locus at which no difference is retained across the declared structure; equivalently, with the fields constant on each connected component. The characterisation is machine-checked over the rationals without classical axioms. Two clarifications carry much of the philosophical weight. The claim is not a redefinition of arithmetic zero, which remains an ordinary rational and an ordinary constructor of number; we mark the arithmetic value and its fibre with different symbols throughout. And the class is relative to the declared structure of comparison rather than absolute, as an example shows in which the operator returns zero on a field that is neither empty nor pointwise zero, and a second in which it returns zero on a field that is not constant. We locate the position against Frege, structuralism, Spencer-Brown’s unmarked state, pseudometric quotients, operational equivalence, and the standard kernel result for graph Laplacians, stating in each case what is borrowed and what is not.

1. The question

Zero is not one question but several, and they are usually run together.

There is a question about the arithmetic object: what kind of thing is the number 0, and what makes statements about it true? Frege answered by way of cardinality [3], set theory by way of the empty set, structuralism by way of position in a pattern [1, 6]. These are live and long-argued positions, and nothing here is offered as a rival to them.

There is a distinct question, less often separated out, about zero as an *output*. An instrument, a computation, a comparison operator—anything that reads a situation and returns a number—may return zero. What has it thereby established? The answer cannot be read off the arithmetic. Knowing what kind of object 0 is does not tell you what a device claims when it hands you one.

The distinction matters most where *difference* is primitive rather than derived. Suppose the ground notion is that two things are told apart, under finite access, with a finite record retained. Number is then generated rather than assumed, and every quantity a reader obtains is the record of some distinction. Zero occupies an odd position in such a setting. It arrives in the same register as every other value—a rational like any other—yet what it reports is that the primitive did not fire. It is the one output that is not the record of a distinction.

This paper asks what that output certifies, and answers with a theorem rather than a stipulation.

2. The setting

Three commitments fix the setting. They are stated here in the form the argument uses; nothing below depends on any further development of them.

The primitive. The ground notion is the *retained difference*: that two things are told apart, with a finite record. Number is generated from a ground together with a successor read as “retain one more distinction”. Nothing in the object language is assumed beneath the primitive.

The reading contract. Every quantity a reader obtains is a rational, delivered at a declared finite resolution. Completed infinities, the real line as a primitive, and the point of zero extent are not in the range of any reading. This constrains what may be emitted; it is not a claim about what exists. One consequence is used repeatedly below: a difference lying outside the range of every admissible reading is not something the setting can assert.

The verdict type. A reader at a finite resolution may fail to determine anything. Readings therefore take values in a type carrying, besides the signed values and zero, a distinct least element $\perp$ for the unresolved case.

That $\perp$ carries no numeric field is a typing constraint rather than a convention: no path extracts a number from an unresolved reading. Section 6 argues that $\perp$ and zero are absences of different kinds and must not be identified.

The comparison operator. Distinctions are compared across a *declared structure*: a finite weighted graph g whose edges (i, j, w) state which pairs are to be compared and with what weight, together with a field Φ assigning a rational to each vertex. The retained information of Φ is

$$I_g(\Phi) = \sum_{(i,j,w) \in g} w(\Phi_i - \Phi_j)^2. \quad (1)$$

The graph is not a picture of the world. It is a statement of which distinctions this reader is arranged to retain. Two vertices joined by no path are two things the reader was never set up to compare—a point that becomes load-bearing in §5.

Remark 1 (The operator is a Dirichlet form). Writing $L_R = D_W - W$ for the graph Laplacian, a one-line ring identity per edge gives

$$w\Phi_i^2 + w\Phi_j^2 - 2w\Phi_i\Phi_j = w(\Phi_i - \Phi_j)^2,$$

and summing over the edge list yields $\Phi^\top L_R \Phi = I_g(\Phi)$. So (1) is the Dirichlet form of L_R , and everything below about I_g is equally a statement about that operator. Nothing in the argument requires this identification; it is recorded because it locates (1) in familiar territory.

3. The kernel

Equation (1) says what the operator computes. The question of §1 concerns what it means when the computation returns zero, and that is a question about the operator's zero locus.

We work over $\mathbb{Q}$ throughout; no completion and no limit appears, and equality on $\mathbb{Q}$ is decidable. Strict positivity of the weights is assumed in the theorems and is not cosmetic: an edge of weight zero contributes nothing to I_g regardless of the field, so a single such edge would let a genuine distinction hide inside the zero locus.

Lemma 2 (Vanishing square). *For $x \in \mathbb{Q}$, if $x \cdot x = 0$ then $x = 0$.*

Proof. $\mathbb{Q}$ is an integral domain, so the product vanishing yields the disjunction $x = 0$ or $x = 0$; both disjuncts are the same statement. $\square$

Lemma 3 (Vanishing sum of nonnegatives). *For $a, b \in \mathbb{Q}$ with $a, b \geq 0$ and $a + b = 0$, both $a = 0$ and $b = 0$.*

Proof. From $b \geq 0$, $a = a + 0 \leq a + b = 0$; with $a \geq 0$, antisymmetry gives $a = 0$, and substitution gives $b = 0$. $\square$

Lemma 4 (Edge criterion). *For an edge (i, j, w) with $w > 0$: the edge term $w(\Phi_i - \Phi_j)^2$ vanishes if and only if $\Phi_i = \Phi_j$.*

Proof. If the product vanishes then by integrality either $w = 0$, excluded, or the square vanishes, whence $\Phi_i = \Phi_j$ by Lemma 2. The converse is immediate. $\square$

Lemma 5 (Termwise vanishing). *If every weight of g is nonnegative and $I_g(\Phi) = 0$, then every edge term vanishes.*

Proof. Induction on the edge list. Each term is a non-negative weight times a square, hence nonnegative, as is the sum over the tail; Lemma 3 splits the vanishing sum at the head, and the tail case is the induction hypothesis. $\square$

Remark 6. Lemma 5 assumes only $w \geq 0$, weaker than the $w > 0$ used below. The separation is deliberate: termwise vanishing needs only nonnegativity, and strict positivity is spent at exactly one point—Lemma 4—to prevent a vanishing weight from absorbing a real distinction.

Theorem 7 (Kernel, edge form). *Let every weight of g be strictly positive. Then for every field Φ ,*

$$I_g(\Phi) = 0 \iff \Phi_i = \Phi_j \text{ for every edge } (i, j, w) \text{ of } g.$$

Proof. $(\Rightarrow)$ By Lemma 5, every edge term vanishes; Lemma 4 converts each to $\Phi_i = \Phi_j$. $(\Leftarrow)$ Induction on the edge list: the head term vanishes by Lemma 4, reducing the sum to the tail, to which the hypothesis applies. $\square$

Let reach_g be the reflexive–transitive closure of the relation “some edge of g joins i and j in either orientation”, defined inductively.

Theorem 8 (Kernel, component form). *Under the same hypothesis, for every field Φ ,*

$$I_g(\Phi) = 0 \iff \Phi_i = \Phi_j \text{ whenever } \text{reach}_g(i, j),$$

that is, if and only if Φ is constant on every connected component of g .

Proof. $(\Rightarrow)$ Theorem 7 gives equality across every edge; induct on the derivation of $\text{reach}_g(i, j)$, the reflexive case being trivial and a step through an adjacency using the edge equality, after symmetry if the orientation is reversed, composed with the induction hypothesis. $(\Leftarrow)$ Every edge makes its endpoints reachable in one step, so the hypothesis yields equality across every edge, and Theorem 7 applies. $\square$

Connectedness is nowhere assumed, and the conclusion is accordingly local to components. §5 argues that this is not a limitation but the content.

Finally, what licenses speaking of a class rather than merely a set of fields.

Proposition 9 (The zero locus is a subspace). *Under strict positivity, $\{\Phi : I_g(\Phi) = 0\}$ contains the zero field and is closed under pointwise addition and under scaling by any $c \in \mathbb{Q}$.*

Proof. By Theorem 7 each condition reduces to the corresponding closure property of “ $\Phi_i = \Phi_j$ on every edge”, which holds because equality in $\mathbb{Q}$ is preserved by addition and multiplication. $\square$

Corollary 10 (Indistinguishability is an equivalence). *Define $\Phi \equiv_g \Psi$ to mean $I_g(\Phi - \Psi) = 0$. Then $\equiv_g$ is reflexive, symmetric and transitive.*

Proof. Reflexivity and symmetry follow from Theorem 7 with the ring structure of $\mathbb{Q}$; transitivity follows by writing $\Phi_i - X_i = (\Phi_i - \Psi_i) + (\Psi_i - X_i)$ edgewise and applying Proposition 9. $\square$

It is worth being exact about what Proposition 9 is needed for. The set $I_g^{-1}(0)$ is the *fibre* of the readout over zero; the class $[\Phi]_{\equiv_g}$ is the set of fields indistinguishable from Φ . These coincide at $\Phi = 0$ trivially, since $\Phi - 0 = \Phi$. What the proposition supplies is not that coincidence but the *transitivity* of $\equiv_g$: without closure of the zero locus under addition, $\equiv_g$ would not be an equivalence and its “classes” would classify nothing.

All results of this section are machine-checked in Coq over the rationals, without `Reals` and without classical axioms; `Print Assumptions` reports *Closed under the global context* on each. The development accompanies this paper as supplementary material and is archived at doi:10.5281/zenodo.21665100, with sources at github.com/morrocwi/zero-readout-certifies.

4. Two zeros, marked differently

The position is easy to state badly, and stating it badly produces a category mistake that is fatal on inspection. We mark the two objects with different symbols and keep them apart.

Definition 11 (Arithmetic zero). $0_{\mathbb{Q}}$ is the rational number zero: an element of the readout’s *codomain*, an ordinary member of $\mathbb{Q}$, and—in a setting that generates number from a ground and a successor—also the ground of that generation.

Definition 12 (Retained-distinction zero). $\mathbf{0}_g := I_g^{-1}(\{0_{\mathbb{Q}}\})$ is the set of fields on which the readout returns $0_{\mathbb{Q}}$: an object in the readout’s *domain*, and by Corollary 10 the $\equiv_g$ -class of the zero field.

The claim of this paper is **not** that $0_{\mathbb{Q}}$ is an equivalence class. That would confuse a value in the codomain with its fibre in the domain, and would additionally deny $0_{\mathbb{Q}}$ a status it plainly has. The claim is:

The foundational content of a readout returning $0_{\mathbb{Q}}$ is membership in $\mathbf{0}_g$.

Theorems 7 and 8 are what make $\mathbf{0}_g$ a determinate object rather than a name: they say exactly which fields it contains.

Example 13 (Zero on a field that is nowhere zero). Let $\Phi_i = 17$ at every vertex of a connected g . Then $I_g(\Phi) = 0_{\mathbb{Q}}$. The field is not empty, no value of it is zero, and nothing about it is absent. What is absent is any difference for the declared structure to retain.

Example 13 does considerable work. It shows immediately that the zero of a discrimination readout is not the zero of emptiness: the certified content of $I_g(\Phi) = 0_{\mathbb{Q}}$ is a claim about differences, not about existence, and a fully populated field with a large uniform value satisfies it. Any reading of $\mathbf{0}_g$ as emptiness is thereby unavailable.

Remark 14 (The ground is untouched). In a setting that generates $\mathbb{N}$ from a ground and a successor, $0_{\mathbb{Q}}$ also appears as that ground. That occurrence is a constructor, not a reading, and nothing here bears on it. A reading of these theorems on which the number tower becomes incoherent is a misreading, and none is intended or licensed.

5. Zero is relative to the declared structure

$\mathbf{0}_g$ carries a subscript, and the subscript is not decoration.

Example 15 (Zero across a gap in the declared structure). Let g have two components A and B , with $\Phi = 0$ on A and $\Phi = 17$ on B . Then $I_g(\Phi) = 0_{\mathbb{Q}}$, although Φ is not constant.

Example 15 looks at first like a defect and is in fact the point. The operator is faithful to distinction *relative to the declared structure of comparison*, not absolutely. Where g supplies no path it supplies no comparison, and the reader was never arranged to tell those vertices apart. That I_g returns $0_{\mathbb{Q}}$ there is not a failure of the operator but an accurate report of the reach of the declaration. Any claim of faithfulness below is relative to g and is so qualified.

The general form is straightforward. For a readout M on states write

$$x \sim_M y \iff M(x) = M(y), \quad \mathbf{0}_M := M^{-1}(\{0_{\mathbb{Q}}\}),$$

and when M is linear, $\mathbf{0}_M = \ker M$. Zero relative to M is the class of states M does not separate.

One clarification is needed, and it marks the point at which a superficially similar formulation would be wrong. It is tempting to say that states in 0_M “differ in themselves though M cannot see it”. Under the reading contract of §2 that is not available: a difference outside the range of every admissible reading is not something the setting can assert. What may be asserted is only that M does not separate them, at the resolution declared, with no claim either way about some other reading. The relativity is not a hedge added for safety; it is the reading contract applied at the kernel, and Example 15 is what it looks like in practice.

6. Zero and refusal are different absences

A second object is easily mistaken for zero, and conflating them is a substantive error rather than a terminological one.

Recall the verdict type of §2. Besides the signed values and zero it carries a least element $\perp$ for the case in which the reader, at its declared resolution, determines nothing. The two absences differ in kind:

0_Q	a fact about the states: no difference is retained
$\perp$	a fact about the reader: it cannot tell

Theorems 7 and 8 are statements about the first, and they are biconditionals precisely because the first is determinate. A determinate absence is exactly what an equivalence class can record. No corresponding biconditional is available for $\perp$: it reports nothing about the field, and therefore characterises no set of fields. This asymmetry is not an incidental feature of the present operator but a consequence of what the two absences are about.

The distinction has a consequence outside philosophy, worth stating because it shows the two are not interchangeable in practice. A spurious zero—an uninitialised accumulator, a default return, an exception swallowed into a zero—is indistinguishable at the point of use from a legitimate small result, and by Theorem 7 it asserts something false and specific: that no difference is retained. A spurious refusal is loud. A system that returns a value on its refusal branch converts the second failure into the first, which is why the separation belongs in the type rather than in a convention.

7. Position in the setting

The results above are not an addition beside the commitments of §2 but a join inside them.

commitment	what the kernel supplies
the primitive: retained difference	the zero locus of the operator <i>is</i> the locus where the primitive retains nothing (Thms. 7–8)
generated number	separates the ground 0_Q of the generation from the 0_Q a reading returns (Rem. 14)
the reading contract	0_M is that contract applied at the kernel; no difference is posited outside a reading (§5)
the verdict type	determinate zero admits a biconditional; the unresolved element does not (§6)
the comparison operator	(1) says what I_g is; the theorems say what $I_g^{-1}(0_Q)$ is

The first row is the one that changes something. Before it, the primitive and the operator were connected by interpretation only—the reading that what L_R carries is information rather than length or energy. That reading was an interpretation and could only be declared as such, because nothing derived it. Theorem 8 derives one half: at its bottom value the operator is exactly the failure of the primitive, so no interpretive step is needed to say what $I_g = 0_Q$ means. What remains interpretive—that the nonzero values measure *information* rather than some other quantity monotone in retained difference—is untouched and stays where it was.

8. Relation to neighbouring positions

Each of the following contains part of what is said above. We state in each case what is borrowed and where the account departs.

8.1. Frege

Frege explains number through equinumerosity and obtains zero as the number belonging to a concept under which nothing falls [3]. The resemblance is real: zero is associated with a class. The departure is twofold. Frege begins from sameness of number, the present account from retained difference. And Fregean zero arises from *empty* cardinality, whereas 0_g routinely contains fields that are fully populated with large values—Example 13 is the counterexample to any reading of 0_g as emptiness. Frege’s zero is also an object of arithmetic, which 0_Q remains here. The accounts are not in competition because they answer different questions.

8.2. Structuralism

Structuralism holds that numbers are positions in a pattern rather than objects with intrinsic natures [1, 6]. It is sometimes offered in support of claims that zero “is not really an object”. It cannot support the present position, for a simple reason: structuralism says the

same of every number. It answers “what is zero within the number structure?”; the question here is “what does a zero-valued reading certify about the states read?”. A structuralist may accept the answer given here without alteration.

8.3. Spencer-Brown’s unmarked state

Of the positions surveyed this is the closest in spirit. *Laws of Form* begins with the injunction to draw a distinction and takes the unmarked state as the condition prior to any mark [7]. Anyone reading an account whose primitive is distinction will hear the echo, and it should be acknowledged rather than left implicit.

The departure is structural. For Spencer-Brown the unmarked state is *primitive*: a starting point of the calculus of indication, prior to and presupposed by every mark. Here non-distinction is *derived*—the zero locus of a quantitative operator, characterised biconditionally, conditioned on a hypothesis ($w > 0$) whose necessity is exhibited. The calculus of indication also has no occasion to separate the unmarked state from a reader’s failure to resolve, whereas §6 requires that separation and locates it in the type. The present account is therefore not a quantitative restatement of *Laws of Form*; it is an account of what a quantitative operator’s vanishing certifies, in a setting where a determinate zero and an unresolved value must be told apart.

8.4. Pseudometrics and quotients

Mathematically this is the nearest neighbour. In a pseudometric space $d(x, y) = 0$ is compatible with $x \neq y$, and the standard quotient collapses precisely the points at distance zero; in a quotient vector space every element of the subspace quotiented out becomes the same zero, so the zero of a quotient is already a class rather than a distinguished element [5]. Proposition 9 is an instance of that construction and no priority is claimed over it.

The addition is not the construction but a theorem about it. Quotient theory tells us that identifying points at distance zero yields a metric; it does not tell us what the zero being quotiented by *means*. Theorems 7 and 8 supply that here by showing the zero locus coincides, in both directions, with the locus at which the primitive fails. The honest formulation is that this account is not independent of quotient mathematics: it gives one quotient class a foundational interpretation together with a faithfulness theorem.

8.5. Operational equivalence

Operational approaches in physics define a state as an equivalence class of preparations that no admitted experiment separates. This is the “relative to a readout” clause of §5 in another setting, and the parallel is close. The difference is one of selection. Operational equivalence treats all classes alike: every observation-

ally indistinguishable pair is grouped and no class is privileged. The present account singles out one class, 0_g , and characterises it by a property the others lack—being exactly the locus at which the primitive fails to fire.

8.6. The standard kernel result

Over the reals it is standard that the Dirichlet energy of a function on a weighted graph vanishes iff the function is constant on components, a fact tied to the zero eigenvalue’s multiplicity counting components [2, 4]. Theorems 7 and 8 claim no novelty against that literature.

What is offered differs in kind from a new graph-theoretic fact: the placement of a known kernel result as a *faithfulness condition* on a distinction-based account—the demonstration that the operator’s zero locus and the primitive’s failure locus coincide in both directions, over $\mathbb{Q}$, with the required hypothesis made explicit, and checked without classical axioms.

9. The reader from inside: two levels of composition

Equation (1) is an extensional presentation of a finite reading. It need not be understood as a view from nowhere in which the whole graph is simultaneously present to an external collector. An internal reader may traverse the declared finite edge list one edge at a time, obtain the corresponding contribution, and update a finite accumulator. The list representation supplies an implementation order; the theorem concerns the value delivered by the declared finite computation, not an epistemic privilege attached to that order.

This raises a genuine typing question. Three states are easily collapsed in prose:

$\perp$ (the accumulator has received no input), $\perp$ (a reading failed)

They are not interchangeable. The first is a process state, the second a failure verdict, and the third a determinate quantitative verdict.

The algebraic source of the confusion is now machine-checked in `ReaderTwoLevels.v`. For one operation, if a single element is both a two-sided identity and a two-sided absorber, every two carrier elements are equal. But the restriction is explicitly *for one operation*. A finite reader performs at least two different compositions:

operation	level	boundary behaviour
accumulation	combine contributions within one reading	an initial state is neutral
pipeline sequencing	compose completed reading stages	an unresolved stage is absorbing

A common carrier element can therefore be neutral for the first operation and absorbing for the second without collapse; the formal file provides an explicit witness. That witness establishes algebraic compatibility only. It does not by itself show that “no accumulated contribution yet” and “the reader failed” have the same meaning. The typed refinement keeps them separate: `Init` is the unit of accumulation, `AccFailed` is absorbing within a failed accumulation, and `Unresolved` is absorbing under pipeline sequencing. A resolved zero remains a different constructor and behaves differently: it allows a later resolved stage to proceed, whereas an unresolved verdict stops the pipeline.

The empty-input case then becomes visible as a reading-contract decision rather than a theorem of accumulation alone. A *total* contract may finalise an untouched accumulator as `Resolved(0Q)`: the traversal completed successfully and accumulated no contribution. A *strict* contract may finalise the same untouched process state as $\perp$: no observation was made, so no quantitative verdict is licensed. Both finalisers are formally defined, and the development proves that they disagree on the empty case.

This qualification blocks an overclaim. Accumulation as such does not force zero. Zero is forced only after additional choices: the process is totalised on an empty trace, the result must inhabit the same quantitative codomain as ordinary readings, and the chosen accumulation law has arithmetic zero as its identity. None of this weakens the kernel theorem. Once the operator legitimately returns 0_Q , Theorems 7 and 8 still determine exactly what that returned value certifies. The new separation says only that a process state must not be smuggled into the verdict type and then mistaken for such a return.

10. What is claimed

It is worth separating the claim from what surrounds it, since each surrounding item is standard.

The mathematics of §3 is elementary and is classical over $\mathbb{R}$. The construction of Proposition 9 is the standard quotient. The relativity of §5 has a close analogue in operational physics. The primacy of distinction has a distinguished predecessor in Spencer-Brown. The formalisation is routine.

What is claimed is an answer to the question of §1, and only that: *where the primitive is retained difference and readings are finite and rational, a readout of 0_Q by a positively weighted discrimination operator certifies membership in 0_g , the class of fields across which the declared structure retains no difference.* The certification is exact in both directions, and it is exact because of a theorem rather than a definition.

Four things are explicitly not claimed. Not that 0_Q fails to be a mathematical object; it is a rational, and in

a generated tower it is also the ground. Not that this displaces any account of arithmetic zero; §8 argues it answers a different question from each of them. Not that the operator is faithful absolutely; Example 15 shows it is faithful relative to the declared structure, which is the strongest form available to a reader with a finite declaration. And not that every form of accumulation must begin from arithmetic zero; §9 shows that the empty case depends on the process type and the finalisation contract.

The position can be put in one sentence.

Arithmetic supplies the symbol 0_Q ; the retained-difference theorem determines what a legitimate occurrence of that symbol certifies.

11. Conclusion

An account that takes distinction as primitive faces a question its arithmetic does not answer: what does it mean when the discriminating operator returns nothing? Left unanswered, zero sits in such an account as an anomaly—the one output that is not a record of the primitive, arriving in the same register as those that are.

The answer given here is that a legitimate zero readout is not an absence but a classification. It places the field read into a determinate class, and Theorems 7 and 8 say which: the fields across which the declared structure of comparison retains no difference. The class is relative to that declaration and is not an absolute; the declaration’s reach is visible in the class, as Example 15 shows. And the class is sharply distinct both from the reader’s inability to decide and from the accumulator’s not-yet-started process state.

The two-level analysis closes the remaining implementation gap. Accumulation within one reading and sequencing across reading stages are different operations, so their boundary elements need not have the same algebraic role. More importantly, an initial accumulator, an unresolved verdict, and a resolved zero are different typed states. Whether an empty traversal finalises as zero or refusal is fixed by the reading contract. Once zero is legitimately returned, however, the kernel theorem determines its content exactly.

The result therefore closes a gap previously bridged by interpretation without claiming that zero is unavoidable in every computational architecture. At the operator’s bottom value the zero locus and the primitive’s failure locus coincide, provably. What remains interpretive—that the nonzero values measure *information* rather than some other quantity monotone in retained difference—is untouched and stays where it was.

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